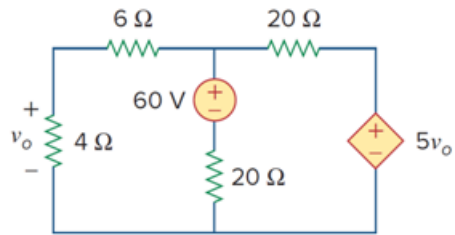


Problem

Using nodal analysis, find v_o in the circuit of Fig. 3.57.

Figure 3.57:



Step-by-step solution

Step 1 of 3

Refer to Figure 3.57 in the textbook.

Label the circuit with the node voltage v_1 .

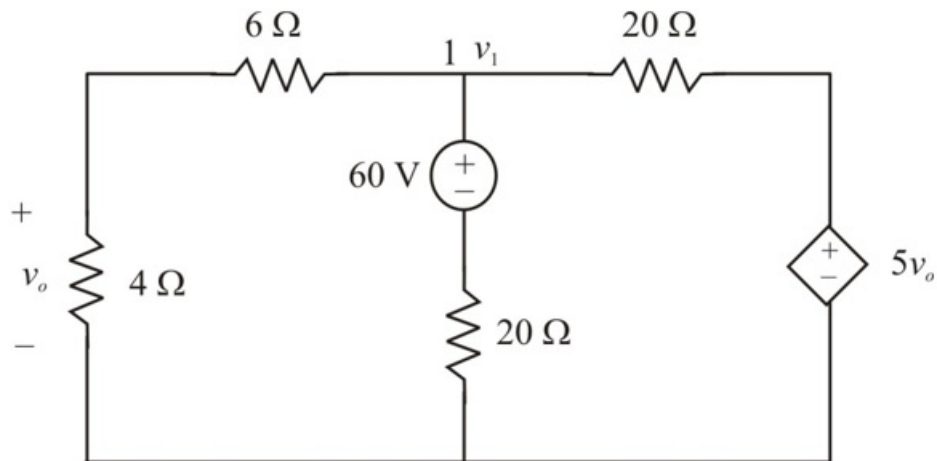


Figure 1

Step 2 of 3

Apply voltage division method to find the voltage v_o from the node voltage, v_1 .

$$\begin{aligned}v_o &= \left(\frac{4}{4+6} \right) v_1 \\&= \left(\frac{4}{10} \right) v_1 \\&= \left(\frac{2}{5} \right) v_1\end{aligned}$$

Write the nodal equation at node 1.

$$\begin{aligned}\frac{v_1 - 0}{6+4} + \frac{v_1 - 60}{20} + \frac{v_1 - 5v_o}{20} &= 0 \\ \frac{v_1}{10} + \frac{v_1 - 60}{20} + \frac{v_1 - 5v_o}{20} &= 0 \\ 2v_1 + v_1 - 60 + v_1 - 5v_o &= 0 \\ 4v_1 - 5v_o &= 60\end{aligned}$$

Step 3 of 3

Substitute the expression of v_o in the nodal equation.

$$\begin{aligned}4v_1 - 5v_o &= 60 \\ 4v_1 - 5\left(\frac{2}{5}v_1\right) &= 60 \\ 4v_1 - 2v_1 &= 60 \\ 2v_1 &= 60 \\ v_1 &= \frac{60}{2} \\ &= 30 \text{ V}\end{aligned}$$

Calculate the voltage, v_o .

$$\begin{aligned}v_o &= \left(\frac{2}{5} \right) v_1 \\&= \left(\frac{2}{5} \right) (30) \\&= 12 \text{ V}\end{aligned}$$

Thus, the voltage v_o is 12 V.